

- 5.3** Show that the characteristic equation for the complement output of a JK flip-flop is

$$Q'(t+1) = J'Q' + KQ$$

- 5.6** A sequential circuit with two D flip-flops A and B , two inputs, x and y ; and one output z is specified by the following next-state and output equations (HDL—see Problem 5.35):

$$A(t+1) = xy' + xB$$

$$B(t+1) = xA + xB'$$

$$z = A$$

- Draw the logic diagram of the circuit.
- List the state table for the sequential circuit.
- Draw the corresponding state diagram.

- 5.8*** Derive the state table and the state diagram of the sequential circuit shown in Fig. P5.8. Explain the function that the circuit performs. (HDL—see Problem 5.36.)

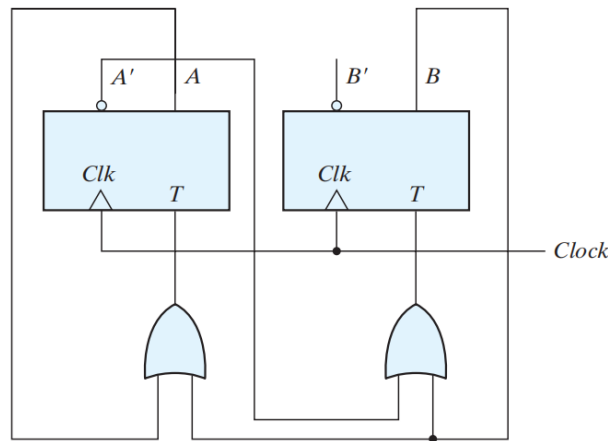


FIGURE P5.8

- 5.9** A sequential circuit has two JK flip-flops A and B and one input x . The circuit is described by the following flip-flop input equations:

$$J_A = x \quad K_A = B$$

$$J_B = x \quad K_B = A'$$

- Derive the state equations $A(t+1)$ and $B(t+1)$ by substituting the input equations for the J and K variables.
- Draw the state diagram of the circuit.

- 5.10** A sequential circuit has two JK flip-flops A and B , two inputs x and y , and one output z . The flip-flop input equations and circuit output equation are

$$J_A = Bx + B'y' \quad K_A = B'xy'$$

$$J_B = A'x \quad K_B = A + xy'$$

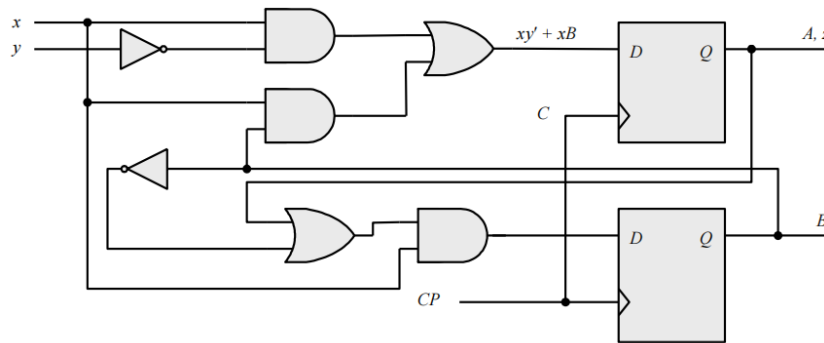
$$z = Ax'y' + Bx'y'$$

- Draw the logic diagram of the circuit.
- Tabulate the state table.
- Derive the state equations for A and B .

5.8

$$Q'(t+1) = (JQ' + K'Q)' = (J' + Q)(K + Q') = J'Q' + KQ$$

5.6



(b)

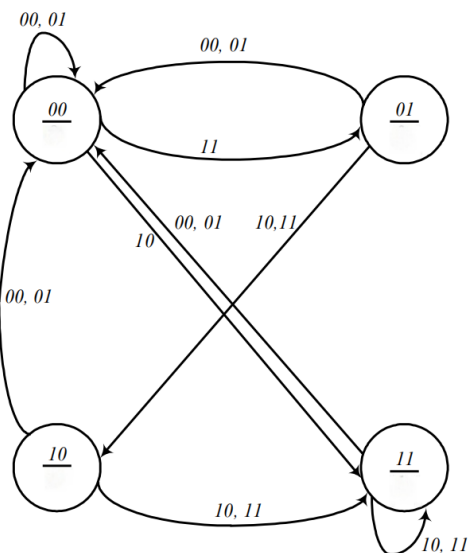
$$A(t+1) = xy' + xB$$

$$B(t+1) = xA + xB'$$

$$z = A$$

Present state		Inputs		Next state		Output
A	B	x	y	A	B	z
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	0	1	0	1	1	0
0	0	1	1	0	1	0
0	1	0	0	0	0	0
0	1	0	1	0	0	0
0	1	1	0	1	0	0
0	1	1	1	1	0	0
1	0	0	0	0	0	1
1	0	0	1	0	0	1
1	0	1	0	1	1	1
1	0	1	1	1	1	1
1	1	0	0	0	0	1
1	1	0	1	0	0	1
1	1	1	0	1	1	1
1	1	1	1	1	1	1

(c)



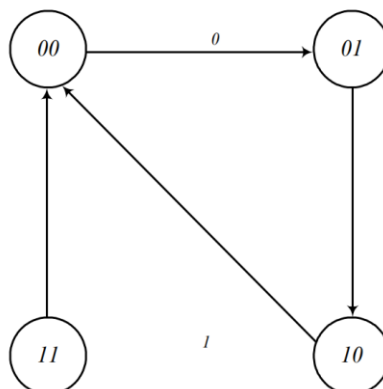
5.8

Present State		Next State		FF Inputs	
A	B	A	B	T_A	T_B
0	0	0	1	0	1
0	1	1	0	1	1
1	0	0	0	1	0
1	1	0	0	1	1

$$T_A = A + B$$

$$T_B = A' + B$$

Repeated sequence: 00 → 01 → 10 → 11 → 00



5.9

$$J_A = x$$

$$K_A = B$$

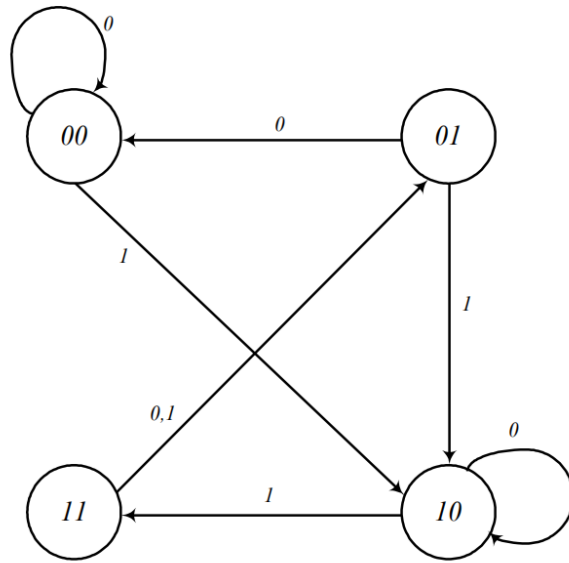
$$J_B = x$$

$$K_B = A'$$

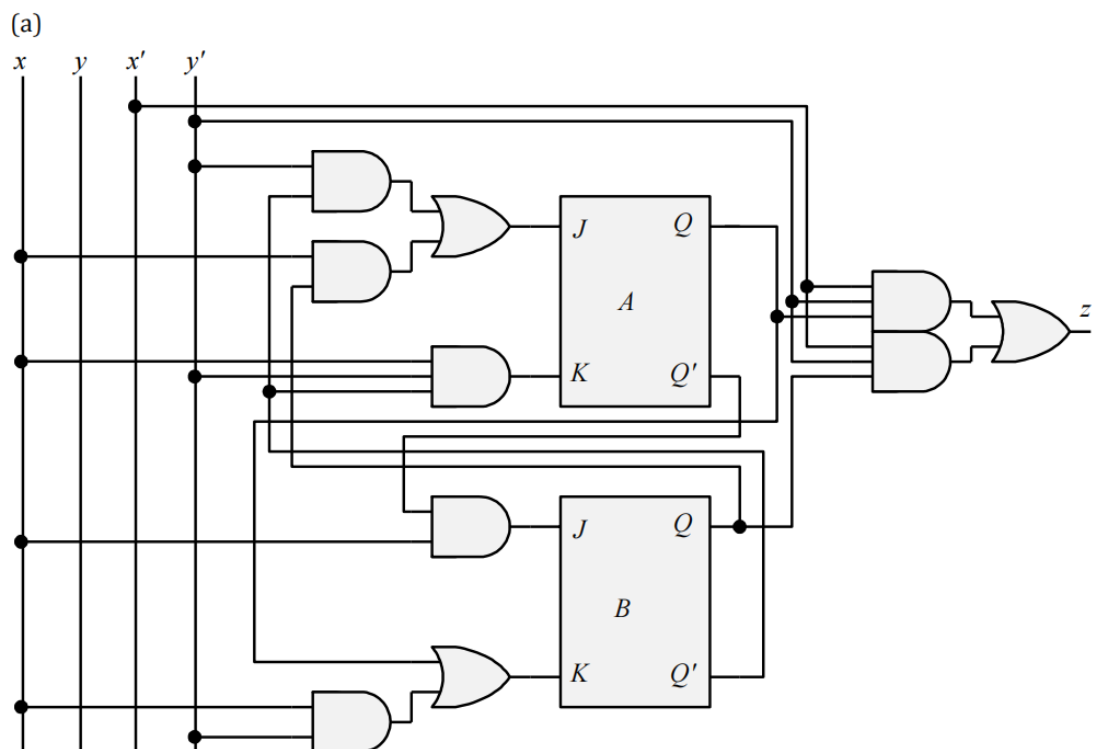
$$A(t+1) = J_A A' + K_A' A = xA' + B'A$$

$$B(t+1) = J_B B' + K_B' B = xB' + AB$$

x	A	B	$xA' + B'A$	$xB' + AB$
0	0	0	0	0
0	0	1	0	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	1
1	0	1	1	0
1	1	0	1	1
1	1	1	0	1



5.10



(b)

<i>Present state</i>		<i>Inputs</i>		<i>Next state</i>		<i>Output</i>	<i>FF Inputs</i>			
<i>A</i>	<i>B</i>	<i>x</i>	<i>y</i>	<i>A</i>	<i>B</i>	<i>z</i>	<i>J_A</i>	<i>K_A</i>	<i>J_B</i>	<i>K_B</i>
0	0	0	0	1	0	0	1	0	0	0
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	1	1	0	1	1	1	1
0	0	1	1	0	1	0	0	0	1	0
0	1	0	0	0	1	1	0	0	0	0
0	1	0	1	0	1	0	0	0	0	0
0	1	1	0	1	0	0	1	0	1	0
0	1	1	1	1	1	0	1	0	1	0
1	0	0	0	1	0	0	1	0	0	1
1	0	0	1	1	0	0	0	0	0	1
1	0	1	0	0	0	0	1	1	0	1
1	0	1	1	1	0	0	0	0	0	1
1	1	0	0	1	0	1	0	0	0	1
1	1	0	1	1	0	0	0	0	0	1
1	1	1	0	1	0	0	1	0	0	1
1	1	1	1	1	0	1	1	0	0	1

(c)

<i>xy</i>		<i>x</i>			
<i>AB</i>		00	01	11	10
<i>A</i>	00	m_0 1	m_1	m_3	m_2 1
	01	m_4	m_5	m_7 1	m_6 1
	11	m_{12} 1	m_{13} 1	m_{15} 1	m_{14} 1
	10	m_8 1	m_9 1	m_{11} 1	m_{10}

$$A(t+1) = Ax' + Bx + Ay + A'B'y'$$

<i>xy</i>		<i>x</i>			
<i>AB</i>		00	01	11	10
<i>A</i>	00	m_0	m_1	m_3 1	m_2 1
	01	m_4 1	m_5 1	m_7 1	m_6
	11	m_{12}	m_{13}	m_{15}	m_{14}
	10	m_8	m_9	m_{11}	m_{10}

$$B(t+1) = A'B'x + A'B'(x' + y)$$